

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second & Fourth Semester of BSc Physics Examination May 2018
(Elective Course)

PD156 Statistics-II

Date:03/05/2018,Thursday

Time:01:30 p.m. to 02:00 p.m.

Maximum Marks:20

MCQ

Important Instructions:

- Tick the correct answer/s and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions.

1. Let A and B be any two events of sample space S , Then the expression for atleast one of the events A and B occur is_____.
(i) $A \cup B$ (iii) $A \cap B'$
(ii) $A \cap B$ (iv) $A' \cap B$
2. If $P(A \cup B) = 0.64$, $P(A \cap B) = 0.16$ and A and B are independent events then $P(A) =$ _____.
(i) 0.8 (iii) 0.25
(ii) 0.48 (iv) 0.4
3. If $P(A) = 0.9$, $P(B) = 0.8$ and $P(A \cap B) = 0.75$ then $P(A \cap B') =$ _____.
(i) 0.95 (iii) 0.18
(ii) 0.15 (iv) 0.1

4. If A_1, A_2 and B are events of a sample space S , then $P(A_1 \cup A_2|B) = \underline{\hspace{2cm}}$.

(i) $P(A_1|B) + P(A_2|B)$

(iii) $P(A_1|B) + P(A_2|B) - P(A_1 \cap A_2|B)$

(ii) $P(A_1|B) - P(A_1 \cap A_2|B)$

(iv) $P(A_1|B)P(A_2|B)$

5. Three events A, B and C are pairwise independent if $\underline{\hspace{2cm}}$.

(i) $P(A \cap B) = P(A)P(B)$

(iii) $P(C \cap B) = P(C)P(B)$

(ii) $P(A \cap C) = P(A)P(C)$

(iv) All (i),(ii) and (iii)

6. If $P(A) = 0.9$ and $P(B) = 0.8$. Then $P(A \cap B) \geq \underline{\hspace{2cm}}$.

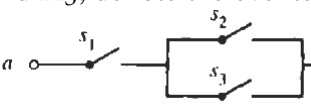
(i) 0.3

(iii) 0.1

(ii) 0.7

(iv) 0.2

7. Let A_1, A_2 and A_3 , denote the events that the switches s_1, s_2 and s_3 are closed,

respectively. . Let A_{ab} , denote the event that there is a closed path between terminals a and b . Then $A_{ab} = \underline{\hspace{2cm}}$.

(i) $A_1 \cup A_2 \cup A_3$

(iii) $A_1 \cap (A_2 \cup A_3)$

(ii) $(A_1 \cap A_2) \cup A_3$

(iv) $A_1 \cap A_2 \cap A_3$

8. If X is the r.v. whose cdf is given by $F(x) = \begin{cases} 0 & x < 0 \\ x + \frac{1}{2} & 0 \leq x < \frac{1}{2} \\ 1 & x \geq \frac{1}{2} \end{cases}$.

Then $P\left(X \leq \frac{1}{4}\right) = \underline{\hspace{2cm}}$

(i) $\frac{3}{4}$

(iii) $\frac{1}{4}$

(ii) $\frac{1}{2}$

(iv) 0

9. Let X be a continuous r.v. X with pdf $f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{Otherwise} \end{cases}$.

Then $P(0 < X \leq 2) = \underline{\hspace{2cm}}$.

(i) $\frac{1}{4}$

(iii) 0

(ii) 1

(iv) 4

10. If $M_X(t)$ is a moment generating function of a random variable X .

Then $E(X) = \underline{\hspace{2cm}}$.

(i) $M'_X(0)$

(iii) $M''_X(t)$

(ii) $M''_X(0)$

(iv) $M'_X(t)$

11. The moment generating function of a random variable X with pdf $f(x)$, is $\frac{e^t - 1}{t}, t \neq 0$. Then $E(X) = \underline{\hspace{2cm}}$.

(i) 0

(iii) $\frac{1}{2}$

(ii) 1

(iv) 2

12. If the probability density function of Beta distributed random variable X is

$$f(x) = \begin{cases} \frac{1}{12}x^2(1-x), & 0 < x < 1 \\ 0, & \text{Otherwise} \end{cases}$$

Then $E(X) = \underline{\hspace{2cm}}$.

(i) $\frac{2}{5}$

(iii) $\frac{5}{2}$

(ii) $\frac{5}{3}$

(iv) $\frac{3}{5}$

13. If random variable X has pdf $f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b; a, b \in \mathcal{R} \\ 0, & \text{Otherwise} \end{cases}$,

Then $P\{|X - E(X)| \leq k\sqrt{V(X)}\} = \underline{\hspace{2cm}}$.

(i) $\frac{k}{\sqrt{3}}$

(iii) $\frac{\sqrt{3}}{k}$

(ii) $\frac{k}{3}$

(iv) $\frac{\sqrt{3}}{k}$

14. If X is a random variable such that $E(X) = 3, V(X) = 4$, then $P(-2 < X < 8) \geq \underline{\hspace{2cm}}$.

(i) 0.16

(iii) 0.84

(ii) 0.4

(iv) 0.6

15. The Moment generating function of distribution $P(X = x) = \begin{cases} p, & x = 1 \\ q, & x = 0 \\ 0, & \text{Otherwise} \end{cases}$

is _____.

(i) $\frac{1}{q + pe^t}$

(ii) $q + pe^t$

(iii) $p + qe^t$

(iv) $\frac{1}{p + qe^t}$

16. For Exponential distribution of a random variable X which is true? _____.

(i) $E(X) = [V(X)]^2$

(ii) $[E(X)]^2 = V(X)$

(iii) $E(X) = V(X)$

(iv) $E(X) = \sqrt{V(X)}$

17. If the m.g.f of random variable X is $M_X(t) = (\frac{1}{3} + \frac{2}{3}e^t)^5$,
then $P(X = 5) =$ _____.

(i) $\left(\frac{2}{3}\right)^5$

(ii) $\left(\frac{1}{3}\right)^5$

(iii) $\frac{2}{3}$

(iv) $\frac{1}{3}$

18. If random variable X has Poisson distribution with parameter μ , then Moment generating function $M_X(t) =$ _____.

(i) $e^{\mu(e^t - 1)}$

(ii) $e^{\mu}(e^t - 1)$

(iii) $e^{-\mu(e^t - 1)}$

(iv) $e^{-\mu}(e^t - 1)$

19. Let X be the number of heads (successes) in $n = 7$ independent tosses of an unbiased coin. Then $M_X(t) =$ _____.

(i) $(1 + e^t)^7$

(ii) $\frac{1}{128}(1 + e^{-t})^7$

(iii) $\frac{1}{128}(1 + e^t)^7$

(iv) $(1 + e^{-t})^7$

20. The cumulative distribution function of pdf

$$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{Otherwise} \end{cases}, a, b \in \mathcal{R} \quad \text{is} \text{_____}.$$

(i) $\frac{x-b}{b-a}$

(ii) $\frac{x-a}{b-a}$

(iii) $\frac{b-a}{x-a}$

(iv) $\frac{1}{b-a}$

End of MCQ

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second & Fourth Semester of BSc Physics Examination May 2018
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PD156 Statistics-II

Date: 03/05/2018, Thursday

Time: 02:00 p.m. to 04:30 p.m.

Maximum Marks: 50

Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION I

Q - II Answer the following questions as directed. **[20]**

1. Define the following terms. **[08]**

- | | |
|--|-----------------------------------|
| (a) Axiomatic definition of Probability. | (c) Probability density function. |
| (b) Baye's Theorem (Statement Only). | (d) Probabilit mass function. |

2. Attempt any three questions. **[12]**

(i) An urn contains five balls numbered 1 to 5 of which the first three are black and the last two are gold. A sample of size 2 is drawn without replacement. Let B_1 denote the event that the first ball drawn is black and B_2 denote the event that the second ball drawn is black.

(a) Describe a sample space for the experiment, and exhibit the events B_1, B_2 and $B_1 \cap B_2$.

(b) Find $P[B_1], P[B_2]$ and $P[B_1 \cap B_2]$.

(ii) Consider the experiment of tossing two dice. Assume that each sample point is equally likely. Define the events A, B , and C :

A = "First die results in a 1, 2, or 3."

B = "First die results in a 3, 4, or 5."

C = "The sum of the two faces is 9."

Determine whether the events A, B and C are independent or pairwise independent.

(iii) If X is the number of white balls in a sample of size 2 drawn without replacement from an urn containing 6 balls of which 4 are white, then determine

(a) all possible values of X (b) Probability distribution of X (c) $E(X)$

(iv) Determine the mean and variance of probability distribution

$$P(X = x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & \lambda > 0, \quad x = 0, 1, 2, \dots \\ 0, & \text{Otherwise} \end{cases} \quad \text{(Do not use moment generating function)}$$

SECTION II

Q - III Answer the following questions. [30]

1. Explain in brief [12]

- (i) Chebyshev's Inequality.
- (ii) Weak law of large numbers.
- (iii) Lindeberg-Lévy Central Limit Theorem.

2. Attempt any three questions. [18]

- (i) Show that the moment-generating function of the random variable X having a Normal probability distribution with mean μ and variance σ^2 is given by
$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

- (ii) From a lot of 10 missiles, 4 are selected at random and fired. If the lot contains 3 defective missiles that will not fire, what is the probability that (a) all 4 will fire? (b) at most 2 will not fire?

- (iii) Buses arrive at a specified stop at 15- minutes intervals starting at 7:00 a.m. That is, they arrive at 7:00, 7: 15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7:00 and 7:30 find the probability that he waits (a) less than 5 minutes for a bus. (b) more than 10 minutes for a bus. (c) between 5 minutes and 10 minutes.

- (iv) Obtain the moment generating function of Negative Binomial distribution with parameters r and p and hence find Mean and Variance of distribution.

- (v) Impurities in batches of product of a chemical process reflect a serious problem. It is known that the proportion of impurities X in a batch has the Beta density

$$\text{function } f(x) = \begin{cases} 10(1-x)^9 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$$

- (a) A batch is considered not sellable and then not acceptable if the percentage of impurities exceeds 60%. With the current quality of the process, what is the probability that a batch is not acceptable?
- (b) Determine the mean proportion of impurities in the batch.
- (c) Determine the standard deviation of proportion of impurities in the batch.
- (vi) Suppose that the length of phone calls in minutes is an exponential random variable with parameter $\lambda = \frac{1}{10}$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait (a) more than 10 minutes; (b) between 10 and 20 minutes. (c) Determine the mean and standard deviation of length of phone call.